CSE470: Software Engineering

Layered Architecture

Monolithic Software

- The end product come at end of the process model
- There is no separation of concern (code) of different software components.
- All code may be written in a single file with html, sql queries, logic checking etc.

Problems of Monolithic Software

- No separation between components
- Changing a component affects other components.

  For example – What if I want to change the UI from JavaScript to Angular ?

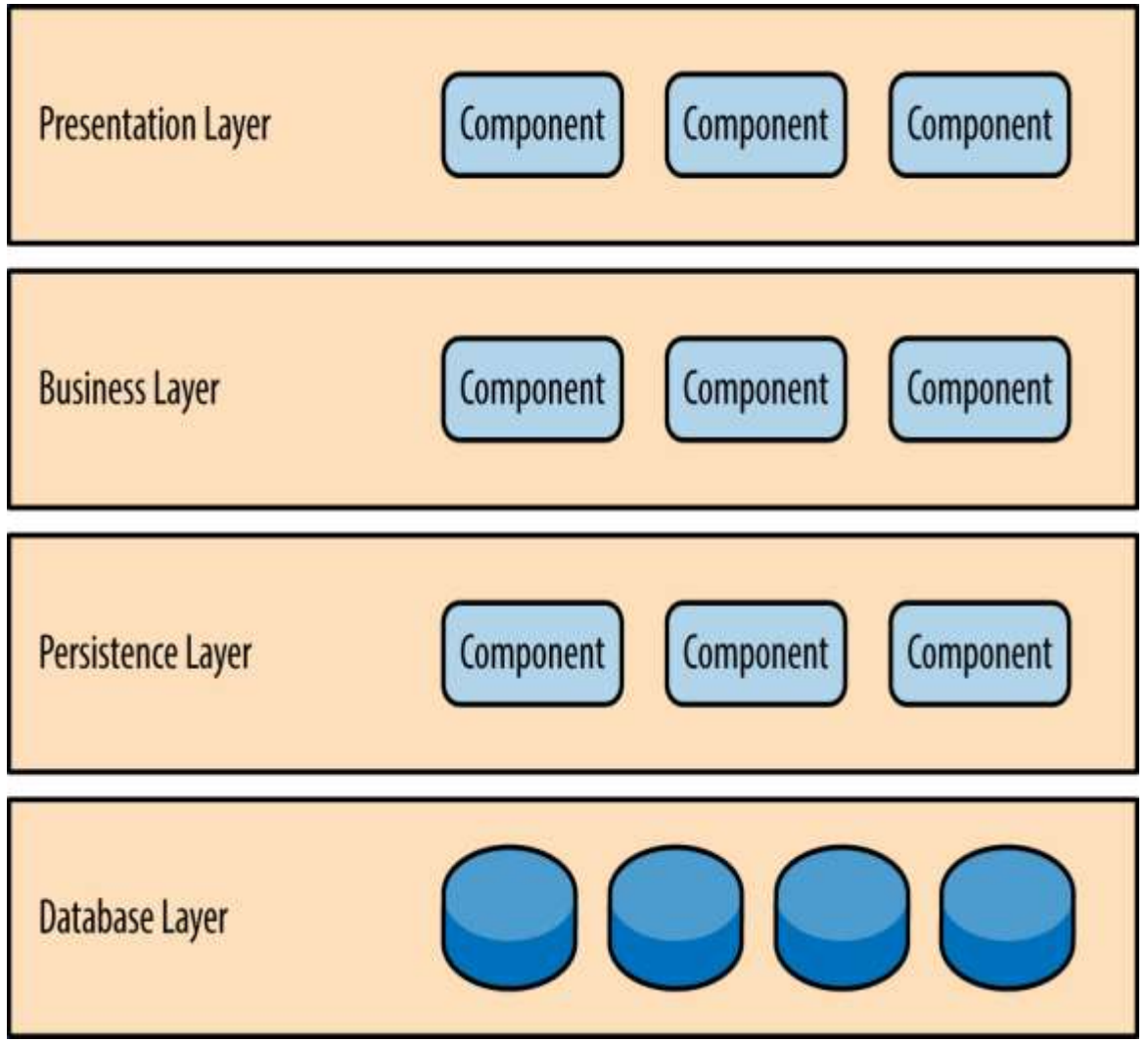
What is Layered Architecture

- Organises the system into a set of layers (or abstract machines) each of which provide a set of services.

- Supports the incremental development of sub-systems in different layers. When a layer interface changes, only the adjacent layer is affected.

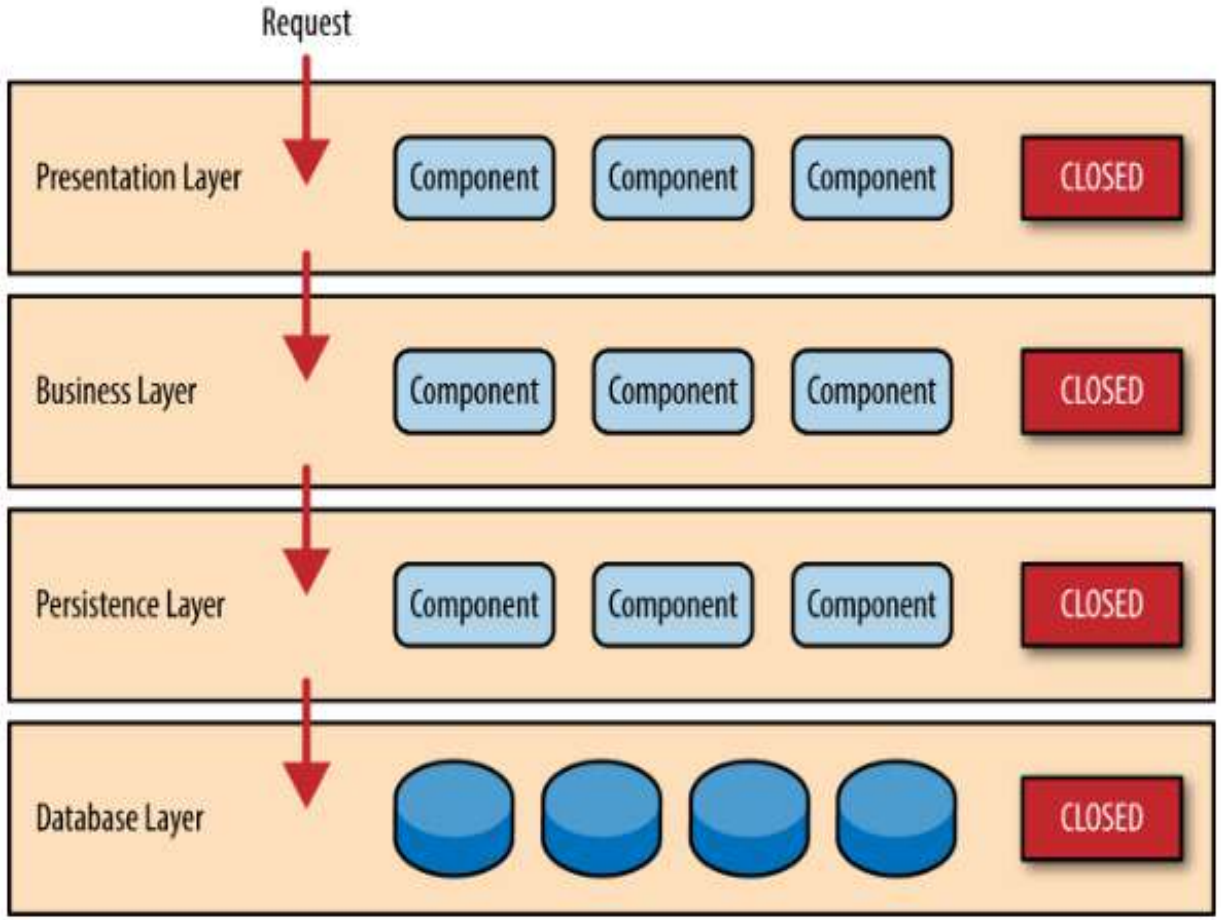
Layered Architecture

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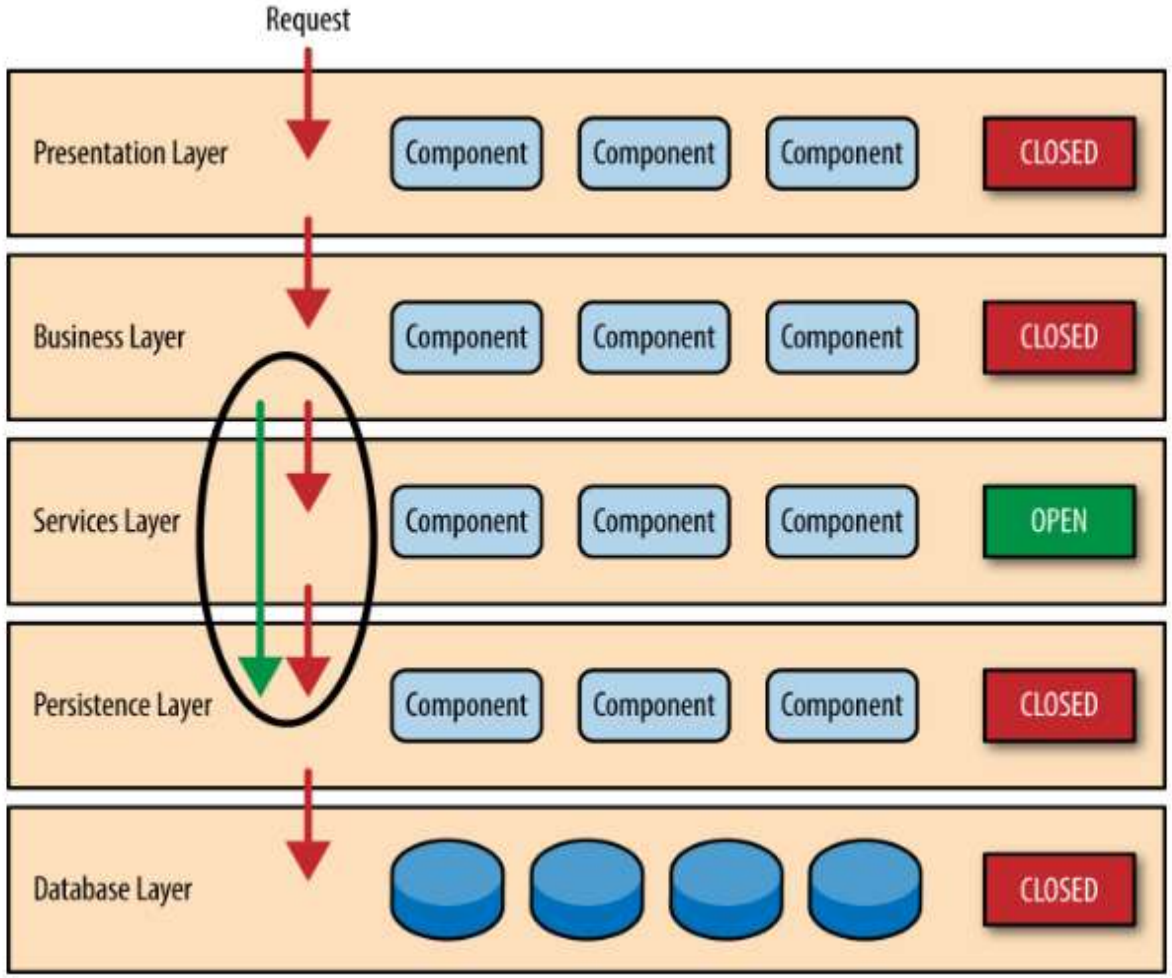
Layered Architecture

- Layers can be open or closed
- A closed layer can only be accessed by the layer above.
- A change in one layer does not affect others. It provides isolation.
- However, what if we want to add a new layer where shared utilities will be provided to be used by the Business Layer. But, we need to use it sometimes.

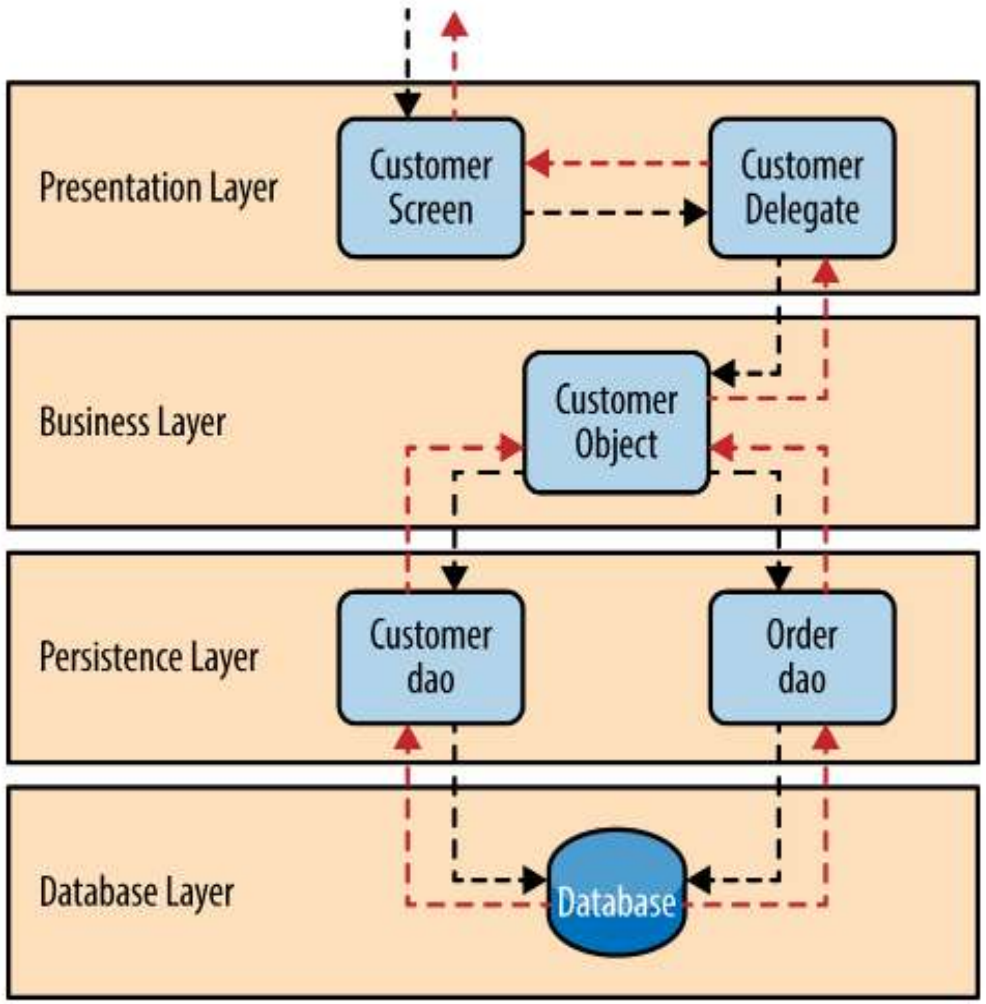


Layered Architecture

- Here comes the concept of open layers.
- An open can be bypassed by upper layers.
- Too many open layers may affect the actual essence of layered architecture.



Example


Presentation Layer
Customer Screen
Customer Delegate
Business Layer
Customer Object
Persistence Layer
Customer dao
Order dao
Database Layer
Database

When Used

- Used when building new facilities on top of existing systems
- When the development is spread across several teams with each team responsible for a layer of functionality
- When there is a requirement for multi-level security (for instance, any secured transaction based system)

Advantages-Disadvantages

Advantages

- Allows easy replacement or addition of entire layers so long as the interface is maintained.
- Testing is easy as components are isolated
- Parallel team work

Disadvantages

- In practice, providing a clean separation between layers is often difficult and a high-level layer may have to interact directly with lower-level layers rather than through the layer immediately below it.
- Performance can be a problem because of multiple levels of interpretation of a service request as it is processed at each layer.
- Parallel processing is not possible (Sequential structure)
- Example: OSI model (7 layers)